

SAKIB SIDDIQUE

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EDUCATION

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

B.Sc. in Civil Engineering; **CGPA: 3.67/4.00**

April 2019 – July 2024

Major: Structural Engineering; **Minor:** Transportation Engineering

Relevant Coursework: Structural Analysis and Design, Design of Reinforced Concrete Structures, Design of Steel Structures, Computer-Aided Analysis and Design of Structures, and Dynamics of Structures.

English Proficiency: IELTS 6.5 (L7, R6.5, W6, S6.5)

SKILLS

Programming & Libraries: MATLAB, C, C++, Python, HTML, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch

Design & Analysis Tools: AutoCAD, ETABS, SAP, SketchUp

Productivity & Management: Microsoft Office, Linux, Adobe Software, Primavera P6

AWARDS & ACHIEVEMENTS

Dean's List Scholarship, BUET: For excellent academic performance.

Technical Scholarship, BUET: Awarded for exceptional achievement in technical courses.

Bangladesh Government Scholarship, Dhaka Board: For academic excellence in the Secondary School Certificate examination.

RESEARCH

International Conference Paper Presented as First Author: Sakib Siddique, [Dr. Shohel Rana](#) (2024)

“A Deep Semi-Supervised Learning based Damage Identification of a Bridge Using Vehicle-Induced Dynamic Displacement and Multilayer Perceptron Model.” IUT Journal of Engineering and Technology ([IUT JET](#)), Islamic University of Technology, Dhaka, Bangladesh. [Archive: IUT JET](#). **[PUBLISHED]**

Bridge Damage Identification Using Bi-LSTM: | Supervisor: [Dr. Shohel Rana](#)

- Developed a Bi-LSTM model to detect and locate structural damage by analyzing differences in vibration data. Applied real-world datasets to improve model accuracy, demonstrating the integration of engineering principles with advanced machine learning techniques *Archive: Undergraduate Thesis, BUET*

Road Damage Detection Using Deep Learning (RDD2022 Benchmark) | Supervisor: [Dr. Celia Shahnaz](#)

- Developing a YOLOv8-based multi-national road damage detection and classification framework, benchmarked on the RDD2022 dataset (47,420 road images, 6 countries), targeting improvement over the YOLOv5 baseline (mAP@0.5 = 0.563) reported in Construction and Building Materials (Elsevier, Q1, 2024); *Approved by Dr. Celia Shahnaz; target journal: IEEE Transactions on Intelligent Transportation Systems.*

Facility Management Data Analytics Project (FMUCD Dataset) | Reviewed by: [Dr. Kahyun Jeon](#)

- Conducted exploratory data analysis and quantitative diagnostics on a 3.7-million-record facility maintenance dataset (12 North American universities), producing a reproducible analysis notebook and a research proposal on predictive maintenance classification. *Independently reviewed and positively assessed by Dr. Kahyun Jeon (IIT).*

PROJECTS

▪ **Exploring Climate-Resilient and Inclusive Latrine Superstructure Designs for Rural Communities in Bangladesh**

- Collaborated with BUET research assistant Mrs. Sayma Simi on a funded research project, preparing detailed AutoCAD drawings for inclusive toilet facilities designed to accommodate diverse user needs. Applied civil engineering design principles to enhance accessibility and inclusivity, contributing to improved sanitation solutions in rural communities.

▪ **Floor Area Ratio (FAR) Project for Ehsan Architects LTD**

- Implemented Python-based algorithms to calculate and analyze building density, ensuring compliance with local zoning regulations and contributing to efficient architectural planning. Demonstrated strong technical skills in Python programming and data analysis for practical engineering applications.

▪ **Capstone Project: Rainwater Harvesting in BUET Area and Redevelopment of Nazrul & Ahsanullah Hall Canteen Area (Role: Team Leader)**

- Conducted feasibility studies, surveys, and data collection to evaluate water harvesting potential, while developing alternative design solutions that accounted for geotechnical, structural, environmental, and transportation factors. Oversaw project planning, design, and report preparation, demonstrating strong leadership and problem-solving abilities.

▪ **Tender Competition: Development of a 6-Storey Residential Building (Role: Team Leader)**

- Estimated the Bill of Quantities (BOQ) and prepared tender documents as part of the CE 302 course, successfully submitting a competitive tender bid.

▪ **Design and Analysis of a 15-Storey Residential Building (ETABS 2020)**

- Conducted structural analysis and design in accordance with BNBC 2020 guidelines, performing FEM analysis using ETABS 2020 and designing structural components for a special moment-resisting frame.

▪ **Design and Analysis of a Steel Warehouse (SAP2000)**

- Performed FEM-based structural analysis and design of steel warehouse structures using SAP2000, ensuring full compliance with BNBC 2020 standards.

REFERENCES

Dr. Shohel Rana | [ResearchGate](#)

- Research: Structural health monitoring, vibration analysis, machine learning for damage detection
- Professor, Department of Civil Engineering, BUET
- Research Interest Score: 159.9 | Citations: 183 | h-index: 7

Dr. A F M Saiful Amin | [Google Scholar](#)

- Professor, Department of Civil Engineering, BUET
- Citations: 1262 | h-index: 16 | i10-index: 18

Nishatee Binte Shahid | [Google Scholar](#)

- Assistant Professor, Department of Civil Engineering, BUET
- Citations: 40 | h-index: 02 | i10-index: 02